

Manish Pandey

Curriculum Vitae

m.pandey@math.au.dk | manishpandey0897@gmail.com | [Personal Webpage](#)

RESEARCH INTERESTS

Probability theory; stochastic processes; random graphs; large deviations; associative memory models.

ACADEMIC POSITIONS

Aarhus University <i>Postdoctoral Researcher</i>	Denmark <i>July-2025 - Present</i>
Eindhoven University of Technology (TU/e) <i>Ph.D. in Mathematics; Supervisor: Prof. Remco van der Hofstad</i>	Netherlands <i>Aug 2020 – Mar 2025</i>

EDUCATION

Indian Statistical Institute, Delhi and Kolkata <i>M.Stat. in Statistics (Probability Specialization); Overall: 89.5%</i>	India <i>Jul 2018 – Jun 2020</i>
Indian Statistical Institute, Bangalore <i>B.Math. (Hons.); Overall: 89.4%</i>	India <i>Jul 2015 – May 2018</i>
St. Theresa Senior Secondary School, Haldwani <i>AISSCE (Class XII); Overall: 93%; Science aggregate: 96.34%</i>	India <i>Apr 2014 – Mar 2015</i>

PUBLICATIONS

1. A. Chakrabarty, **M. Pandey**, S. Chakrabarty, *Length of stationary Gaussian excursions*, *Proceedings of the American Mathematical Society* **151**(3): 1339–1348, 2023.
2. R. van der Hofstad, **M. Pandey**, *Connectivity of random graphs after centrality-based vertex removal*, *Journal of Applied Probability* **61**: 967–998, 2024.
3. R. van der Hofstad, **M. Pandey**, *Are giants in random digraphs “almost” local?*, *Electronic Communications in Probability* **30**: 1–13, 2025.
4. O. Nagy, **M. Pandey**, G. Exarchakos, M. Bentum, R. van der Hofstad, *Communication protocol for a satellite-swarm interferometer for low-frequency radio astronomy*, *Acta Astronautica* **246**: 821–831, 2026.
5. M. A. Ramos-Docampo, C. Duan, N. Wang, **M. Pandey**, P. de Dios Andres, C. Hirsch, B. M. Städler, *Modelling active particle motion from fluorescence correlation spectroscopy data*, accepted for publication in *Analytical Chemistry*, 2026.

PREPRINTS AND SUBMITTED MANUSCRIPTS

1. G. Exarchakos, R. van der Hofstad, O. Nagy, **M. Pandey**, *Bringing order to network centrality measures*, *arXiv:2601.16236*, submitted, 2026.

THESIS AND EXPOSITORY WRITING

Ph.D. thesis: *Centrality measures and connectivity properties in large networks*, Eindhoven University of Technology, 2025.

M. Pandey, *Centrality Measures: Who Is the Most Important in a Network?*, *The Network Pages*, 2023.

HONOURS AND AWARDS

- ISI Teachers' Committee Award for outstanding academic performance in the M.Stat. programme (2018–2020).
- ISI Teachers' Committee Award for outstanding academic performance in the B.Math. (Hons.) programme (5 of 6 semesters, 2015–2018).
- Institute stipend/fellowship throughout the M.Stat. (2018–2020) and B.Math. (Hons.) (2015–2018) programmes.
- Nominee for the P. C. Mahalanobis Gold Medal (Top 4 in the M.Stat. cohort, Indian Statistical Institute).
- International Rank 41 and Zonal Rank 1 (Uttar Pradesh and Uttarakhand), SOF International Mathematics Olympiad, 2015.
- All India Rank 9, M.Stat. Entrance Examination, Indian Statistical Institute.
- Top 1% in NSEP (2014–2015); Merit Certificate, IAPT.

RESEARCH VISITS

University of Augsburg

Visiting Researcher

March 2026

Department of Mathematics

Purdue University

Visiting Research Scholar

Summer 2024

School of Industrial Engineering

TEACHING AND SUPERVISION EXPERIENCE

Aarhus University

Lecturer: Random Networks

Spring 2026

Department of Mathematics

- Designed and taught the entire Master's-level course, including lectures, problem sets, and instruction sessions, on random network models and local limits.

Aarhus University

Lecturer: Monte Carlo Simulation

Fall 2026

Department of Mathematics

Aarhus University

Co-supervisor of Ph.D. student

Aug 2025 – Present

Department of Mathematics

Eindhoven University of Technology
Teaching Assistant: Calculus, Probability, and Statistics

Aug 2020 – Feb 2024
Department of Mathematics

INVITED TALKS AND PRESENTATIONS

University of Augsburg

Mar 2026

Invited Speaker

Department of Mathematics

Concluding NETWORKS Conference

Oct 2025

Invited Speaker

The Netherlands

ICTS-NETWORKS Workshop “Challenges in Networks”

29 Jan–02 Feb 2024

Poster Presentation

ICTS Bengaluru, India

LMS Probability Research School

26–30 Jun 2023

Poster Presentation

University of Liverpool

NETWORKS Training Week

May 2022; Oct 2022

Speaker

Asperen, the Netherlands

Institut Henri Poincaré, Paris

Jul 2021

Invited Speaker

Bézout–Eurandom Conference / Franco–Dutch Meeting

Master Classes in Probability

Jan 2020

Invited Participant

University of Strasbourg

INTERNSHIPS

State Street Global Advisors

Summer 2019

Active Quantitative Equity Intern

Bangalore, India

NovelSynth Soft Solutions (Petrabytes)

Winter 2017

Research Intern, Machine Learning

Bangalore, India

ONGOING PROJECTS

Cascades in networks; Centrality comparison curves; Hopfield model in geometric random graphs; Locality properties of centrality measures; Large-deviation analysis of sparse many-body systems in motion.

TECHNICAL SKILLS

Programming Languages: Python, C/C++, R, HTML.

Tools and Software: L^AT_EX, MATLAB, Microsoft Office.

LANGUAGES

English; Hindi.